HomeMaster Thermostat for OpenTherm — Datasheet

Part No. OTT-R1 · Hardware V1.0 · Firmware 1.0.0 Document No.: DS-OTT-R1-V1.0 · November 2026

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1. Description

Wall-mounted ESP32 room thermostat that acts as an **OpenTherm master** for modulating boilers. 2.8″ colour display, rotary encoder and four keys, 24 GHz presence radar, on-board temperature, humidity and ambient-light sensors.

The heating control loop — schedule, room hysteresis, weather-compensated flow setpoint — runs on the device. Home Assistant is used for visibility and remote control, not as a dependency: with the network down the room keeps being heated to the stored schedule.

ESPHome is pre-installed. The device appears in Home Assistant over the native ESPHome API, with no MQTT and no register mapping.

2. Key features

- OpenTherm master, full boiler telemetry and modulating control
- Local control loop — no cloud, no internet, no Home Assistant required
- 2.8″ IPS colour display, 320 × 240, ST7789V
- Rotary encoder and four perimeter keys
- HLK-LD2410C 24 GHz presence radar, UART plus instant GPIO output
- SHT40 temperature and humidity sensor on a thermally isolated tab
- BH1750 ambient light sensor
- Improv Wi-Fi onboarding over Bluetooth and USB serial
- OTA updates over HTTP (vendor-managed) and over the ESPHome API
- Open hardware (CERN-OHL-W v2) and firmware (MIT)

3. Absolute maximum ratings

Parameter	Min	Max	Unit
Supply voltage at +V / 0V	−30	+30	V DC
Operating ambient temperature	0	+40	°C
Storage temperature	−20	+70	°C
Relative humidity, non-condensing	0	90	%

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Stresses beyond those listed may cause permanent damage. The supply input is reverse-polarity protected by a series Schottky diode.

4. Electrical characteristics

Parameter	Value
Nominal supply	24 V DC, SELV
Supply current, measured across operating states	50–100 mA
Input power, maximum	2.4 W
Energy source classification	ES1 throughout; PS1
Equipment class	Class III, no protective earth
OpenTherm interface	Master, 2-wire, polarity-insensitive
OpenTherm cable length	50 m max., twisted pair
USB	USB-C, serial only, CP2102

Current and power figures are those measured on the assembled unit at 24 V DC with the backlight at 100 % and both radios active, as recorded in safety assessment SAFETY-OTT-R1-2026-001.

5. Radio

	Wi-Fi / Bluetooth	Presence radar
Module	ESP32-WROOM-32U-N16	HLK-LD2410C
Band	2402–2480 MHz	24.00–24.25 GHz
Max. radiated power	≤ 20 dBm EIRP	−5.70 dBm e.i.r.p., measured
Antenna	External, U.FL connector	On-module
Standards	802.11 b/g/n, BLE	FMCW presence detection

Detection range of the radar is configurable per gate from Home Assistant (**Radar Max Move Gate**, **Radar Max Still Gate**, **Radar Timeout**).

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6. Display and controls

Item	Value
Panel	2.8″ IPS, 320 × 240, ST7789V, SPI
Backlight	LEDC PWM at 25 kHz, above the audible range
Backlight level, awake	85 %
Encoder	Quadrature, 2 counts per detent
Keys	Four: ◀ ▶ ▲ ▼
Screen refresh	Every 2 s, and immediately on any input
Screen blanking	Radar-driven, 45 s after the room empties

Four pages — Climate, Boiler, Schedule, System — each with a menu reached with ▲ or ▼. The interface is described in full in Chapter 4 of the user manual.

7. Sensors

Sensor	Part	Interface	Address
Temperature and humidity	SHT40	I²C	0x44
Ambient light	BH1750	I²C	0x23

8. GPIO map

Function	GPIO
OpenTherm in / out	21 / 26
Display SCK / MOSI	5 / 23
Display CS / DC / RST	17 / 19 / 13
Display backlight PWM	25
I²C SDA / SCL	18 / 22
Radar UART TX / RX	14 / 16
Radar presence output	4
Encoder A / B	32 / 33
Key ▶ / ▲ / ▼ / ◀	34 / 35 / 36 / 39

GPIO5 is a strapping pin, used here as the display clock. It is driven only after boot and carries no external pull resistor. GPIO34/35/36/39 are input-only; each key has an external 10 kΩ pull-up to 3V3 and a 100 nF debounce capacitor.

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9. Terminals

Terminal	Function
+V	24 V DC supply
0V	0 V
OT+	OpenTherm bus, boiler terminal 1
OT−	OpenTherm bus, boiler terminal 2

10. Mechanical

Enclosure: Italtronic EYE BOX, ABS to UL94 V-0, white (RAL 9003), wall mounted.

Dimension	Value	Italtronic part
Body outer diameter	Ø 94.6 mm	P61610001G
Front aperture	Ø 82.2 mm	P61610001G
Ring height	20.5 mm	P61610001G
Wall base, round	Ø 88.7 mm, 10.5 mm deep, keyhole centres 58 mm	P61610002B
Wall plate, rectangular	140 × 80 mm, 9 mm thick, aperture 45 × 30 mm, fixings 83.5 / 60 mm	P61610014A
PCB carrier	Ø 90.3 mm, 7.5 mm deep; PCB holes 58 × 36 mm, Ø 2.4 mm	P61610001V

Assembled unit: base plate 152 × 107 mm, round body Ø 95 mm, total height from base to screen 47 mm.

11. Default entity set

Sensors: Room Temperature, Room Temperature Raw, Room Humidity, Ambient Light, Heating Target, Flow Setpoint By Curve, Boiler Water Temperature, Boiler Return Temperature, Boiler DHW Temperature, Boiler CH Pressure, Boiler Relative Modulation Level, Outside Temperature, Uptime, WiFi Signal, Radar distances.

Binary sensors: Boiler Flame On, Boiler CH Active, Boiler DHW Active, Boiler Fault Indication, Boiler Low Water Pressure, Boiler Flame Fault, Boiler Service Request, Comfort Window, Heat Demand, Presence, Radar Target.

Numbers: Comfort Temperature, Eco Temperature, Boost Offset, Room Hysteresis, Room Sensor Offset, Curve Cold/Warm Outside, Curve Cold/Warm Flow, Boiler DHW Setpoint, Boiler Max CH Setpoint, Radar gates and timeout.

Datetimes: Weekday and Weekend Comfort Start and End.

Switches: Heating Automatic, Boost Until Next Change, Boiler CH Enable, Boiler DHW Enable.

Other: Backlight (light), Firmware Update (update), IP Address and ESPHome Version (text sensors).

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12. Defaults

Setting	Default
Comfort Temperature	21.5 °C
Eco Temperature	19.0 °C
Boost Offset	1.0 °C
Room Hysteresis	0.3 °C
Weekday comfort window	17:00 – 23:00
Weekend comfort window	09:00 – 23:00
Weather curve	−15 °C → 65 °C, +15 °C → 35 °C
Control interval	30 s
Screen timeout	45 s

13. Compliance

Directive 2014/53/EU (RED), Directive 2011/65/EU and (EU) 2015/863 (RoHS), Directive 2012/19/EU (WEEE), Regulation (EU) 2023/988 (GPSR). See DoC-OTT-R1-V1.0 and Chapter 12 of the user manual.

14. Ordering information

Part number	Description
OTT-R1	HomeMaster Thermostat for OpenTherm, wall mount, 24 V DC

15. Manufacturer

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